

Suryansh Singh Rawat

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EDUCATION

University of Washington, Seattle

Sept 2025 – Mar 2027

Master of Science (M.S), Data Science | **GPA: 3.81/4.0**

Seattle, WA

- Relevant Courses: Natural Language Processing, Machine Learning, Probability and Statistics, Applied Statistics and Experimental Design, Software Design for Data Science, Scalable Data Systems and Algorithms

Birla Institute of Technology and Science, Pilani

Aug 2018 – July 2022

Bachelor of Engineering (B.E), Electronics and Instrumentation

Goa, India

TECHNICAL SKILLS

Languages: Python, C++, SQL, R, Java, TypeScript, Bash, MATLAB

Developer Tools: Git, AWS, Microsoft Azure, Docker, S3, DynamoDB, PostgreSQL, MongoDB, Redis, Supabase

Frameworks: PyTorch, TensorFlow, Keras, SpaCy, HuggingFace, CUDA, LangChain, LangGraph, Streamlit, FastAPI, Selenium

WORK EXPERIENCE

Ascentt

Mar 2024 – Sept 2025

Principal AI Engineer

Remote, US

- Scaled ToyotaGPT (ChatGPT equivalent for enterprises) to **74,000+** active users, architecting **REST APIs**, **DynamoDB** for user state management, and an **Agentic RAG** pipeline using **LangGraph** managing **1TB+ of unstructured data** across **14+** document types.
- Optimized RAG pipelines across **4 enterprise** products, implementing lossless preprocessing, chunking, and indexing in **Amazon OpenSearch**, and benchmarking with **RAGAS**, achieving **+25% Precision@10** and **+12% Recall@10** over vanilla RAG baseline.
- Owned secure file ingestion for enterprise AI systems, designing **encrypted upload** and preprocessing pipelines using **AES-GCM with AWS KMS**, enabling compliant handling of sensitive documents across multiple file types.
- Led backend repository maintenance** and code reviews, ensuring high-quality, production-ready code across enterprise systems.

Deloitte USI

July 2022 – Mar 2024

Data Scientist | AI Center of Excellence Team

Bangalore, India

- Developed **Autoencoder anomaly detection** models with graph-based preprocessing in **Neo4J** to detect **zero-day threat attacks** from real-time AWS network flow data.
- Researched an NLP pipeline for regulatory document consolidation, leveraging **metric learning** trained **SecBERT** embeddings and **HDBSCAN** to merge semantically overlapping compliance requirements, achieving **~83% reduction** in manual SME efforts.
- Constructed a **SecBERT text summarization model** for generating cyber threat reports, reducing analyst review time by **~40%**, integrating **NER tagging and masking** with **SpaCy** annotation, deploying end-to-end pipeline using **Flask and Docker**.

Scienaptic AI

July 2021 – Dec 2021

Data Scientist Intern

Bangalore, India

- Achieved an **11.5% increase** in application approvals by building credit underwriting models using **XGBoost and Logistic Regression** to forecast the probability of credit accounts that may default.
- Built a unified credit underwriting model using **100M+ raw credit records** to deliver baseline metrics across diverse risk segments.

SELECTED PROJECTS

OpenProbe 🔗 | Python, Langchain, LangGraph

May 2025 – June 2025

- Built OpenProbe, a deep research **multi agent orchestration** using **LangGraph** state orchestration (websearch, code, reasoning).
- Achieved **67.1% accuracy** on the FRAMES benchmark, outperforming GPT-4o-Search by **1.5%**, using DeepSeek-R1/Qwen3-32B.

Ad-Astra: xAI Ad Intelligence Engine 🔗 | Python, Grok, XAPI, Supabase

Dec 2025 – Dec 2025

- Built a personalized ad generation and placement engine for X (Twitter), using **Grok 4.1 Reasoning & Vision models** for persona-aware ad copy generation and **scroll telemetry** to optimize in-feed placement; improving engagement and ad relevance.

PersonaAI 🔗 | Python, Microsoft Phi-4, Supabase

Oct 2025 – Oct 2025

- Built AI platform **simulating A/B tests** on **1M+ NVIDIA Nemotron Personas** dataset, to predict digital campaign performance.
- Created natural-language audience targeting with variant-level diagnostics; **3rd place at Lovable Hackathon** against **40+** teams.

KeyCognition 🔗 | Python, PyTorch, React

Nov 2025 – Nov 2025

- Built a behavioural analysis platform processing **136M+ keystroke events** to predict user cognitive load (low/med/high).
- Leveraged temporal typing features (e.g., spelling mistakes, backspaces) with **LSTM and K-Means models** for user behavior classification, winning **3rd place at the UW Databricks Hackathon**.

Detecting GAN Generated DeepFake Images 🔗 | Python, Tensorflow, Deep Learning

Jan 2021 – June 2021

- Designed lightweight CNN architecture to detect **GAN-generated** DeepFakes, achieved **97.77% accuracy** on StyleGAN dataset with SOTA-level precision and recall.